

Austin J. Ferguson

ferguaus@gvsu.edu | <https://fergusonaj.github.io/>
Mackinac Hall C-2-216, 1 Campus Dr | Allendale, MI 49401

Education

Dual Ph.D. Computer Science & Engineering; Ecology, Evolution, and Behavior

Advisor: Dr. Charles Ofria

Dissertation Title: An Empirical Analysis of Historical Contingency in Evolution

Michigan State University, East Lansing, Michigan

August 2018 - June 2024

Dual B.S. Computer Engineering Technology; Digital Simulation & Gaming Engineering Technology

Minor in Mathematical Science - Summa Cum Laude

Shawnee State University, Portsmouth, Ohio

August 2014 - May 2018

Professional Appointments

Assistant Professor - Department of Computer Science, College of Computing

Grand Valley State University, Allendale, Michigan

August 2024 - Present

Peer-reviewed Journal Articles

Alexander Lalejini, **Austin J. Ferguson**, Nkrumah A. Grant, and Charles Ofria (2021). [Adaptive Phenotypic Plasticity Stabilizes Evolution in Fluctuating Environments](#). *Frontiers in Ecology and Evolution*, Vol. 9.

Peer-reviewed Conference Publications

Nikolai Escondo, **Austin J. Ferguson** (2026). [Summarizing Populations: Characterizing the Effects of Sampling in Computational Evolutionary Replay Experiments](#). *Proceedings of the 2026 Conference on Artificial Life*, 88-97.

Marcos Sanson, **Austin J. Ferguson** (2025). [A computational baseline for changes in long-term evolvability](#). *Proceedings of the 2025 Conference on Artificial Life*, 623-632.

Grant Gordon, **Austin J. Ferguson**, Emily Dolson, and Alexander Lalejini (2025). [Environmental connectivity influences long-term evolutionary outcomes](#). *Proceedings of the 2025 Conference on Artificial Life*, 313-322.

Austin J. Ferguson, Clifford Bohm, and Charles Ofria (2024). [Predicting the Unpredictable: Using replay experiments to disentangle how evolutionary outcomes are altered by adaptive momentum](#). *Proceedings of the 2024 Conference on Artificial Life*, 69-77.

Austin J. Ferguson and Charles Ofria (2023). [Potentiating Mutations Facilitate the Evolution of Associative Learning in Digital Organisms](#). *Proceedings of the 2023 Conference on Artificial Life*, 496-504.

Katherine G. Skocelas, **Austin J. Ferguson**, Clifford Bohm, Katherine Perry, Rosemary Adaji, Charles Ofria (2022). [The Evolution of Genetic Robustness for Cellular Cooperation in Early Multicellular Organisms](#). *Proceedings of the 2022 Conference on Artificial Life*, 345-353.

Alexander Lalejini, Emily Dolson, Clifford Bohm, **Austin J. Ferguson**, David P. Parsons, Penelope Faulkner Rainford, Paul Richmond, and Charles Ofria (2019). [Data Standards for Artificial Life Software](#). *Proceedings of the 2019 Conference on Artificial Life*, 507-514.

Peer-reviewed Extended Abstracts

- Alexander Lalejini, **Austin J. Ferguson**, Nkrumah A. Grant, and Charles Ofria (2022). [The evolution of adaptive phenotypic plasticity stabilizes populations against environmental fluctuations](#). (Article summary). Proceedings of the 2022 Conference on Artificial Life, 143-145.
- Acacia Ackles, **Austin J. Ferguson**, Connor Grady, and Charles Ofria (2020). [Rank-based epistasis: A new metric for understanding epistasis in the absence of quantifiable fitness interactions](#). Proceedings of the 2020 Conference on Artificial Life, 160-162.

Book Chapters

- Austin J. Ferguson**, Alexander Lalejini (2027 - In Press). Identifying potentiating events in evolutionary search using replay experiments. Genetic Programming Theory and Practice XXIII.
- Austin J. Ferguson**, Jose Guadalupe Hernandez, Daniel Junghans, Alexander Lalejini, Emily Dolson, and Charles Ofria (2020). [Characterizing the Effects of Random Subsampling on Lexicase Selection](#). Genetic Programming Theory and Practice XVII.

Funding and Fellowships

NSF Research Experience for Undergraduates (REU) ([NSF Award #2547922](#), 2026-2029)

Erik Fredericks (PI), Austin Ferguson (Co-PI).

REU Site: Research Experience for Undergraduates Intersecting Computing and Evolution.

Awarded amount: \$370,039

GVSU Annis Water Resources Institute Faculty Fellow (F2026-W2027)

Award: Two semesters of course release

GVSU Kindschi Undergraduate Research Fellowships (\$3,500 each):

1. Fall 2026 - Nikolai Escondo (undergraduate mentee), Austin Ferguson (PI).
Title: Evolvability Landscapes.
2. Fall 2026 - Marcus Abad-Dominguez (undergraduate mentee), Austin Ferguson (PI).
Title: Computationally modeling the evolution of invasive species undergoing repeated bottlenecks.
3. Winter 2025 - Marcos Sanson (undergraduate mentee), Austin Ferguson (PI).
Title: Identifying Evolvability-Enhancing Mutations via Computational Models.

GVSU College of Computing Seed Funding (2024)

Erik Fredericks (PI), Alexander Lalejini (Co-PI), Austin Ferguson (Co-PI), Byron DeVries (Co-PI).

Research Experience for Undergraduates Intersecting Computing and Evolution.

Awarded amount: \$23,000

BEACON Science and Technology Center Research Grant (2021)

Alternate Timelines: The Role of History in the Digital Evolution of Associative Learning.

Austin Ferguson (graduate student), Charles Ofria (PI) Awarded amount: est. \$45,000

BEACON Science and Technology Center Top-Up Fellowship (2018-2023)

MSU Engineering Distinguished Scholar (2018-2023)

Contributed Talks

Potentiating Mutations Facilitate the Evolution of Associative Learning in Digital Organisms

Authors: Austin J. Ferguson (speaker), Charles Ofria.

Presented at the 2023 Artificial Life Conference

July 2023

Adaptive phenotypic plasticity stabilizes evolution in fluctuating environments.

Authors: Alexander Lalejini, Austin J. Ferguson (speaker), Nkrumah A. Grant, Charles Ofria.

Presented at the 2022 Artificial Life Conference

July 2022

Presented at the ALife 2022 ERA workshop for the Outstanding Student Publication award July 2022

Seminar Presentations

- Replay Experiments Demonstrate Changing Evolutionary Potential due to Adaptive Momentum
Presented at EEB symposium at MSU May 2024
- Historical contingency, potentiation, and digital evolution (oh my!)
Presented at BEACON multi-lab meeting at MSU September 2023
- What is digital evolution and what can it do?
Presented at CSE graduate seminar at MSU April 2023

Workshops

- Co-organizer** - Origins and Transitions in the Evolution of Learning (OTEL) Artificial Life 2026
Brought external speakers to discuss the evolution of learning across taxa and scales with the ALife community.
- Co-organizer** - Emerging Researchers in Artificial Life Workshop ALife 2020-2021, 2023
Provided an opportunity for students, postdocs, and early-career researchers to learn about the group, present their research, and network in a low-stress environment.

Teaching and Mentoring Experience

- Research Mentor** - Mentored the following undergraduates at GVSU:
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|---|---------------|
| Lilianna Zubalik (co-mentored with Alex Lalejini) | S2026 - F2026 |
| Marcus Abad-Dominguez | W2026 - F2026 |
| Nikolai Escondo | F2025 - F2026 |
| Adam Opperman (co-mentored with Alex Lalejini) | Summer 2026 |
| Marcos Sanson | Winter 2025 |
- Instructor** - CIS-162 Computer Science I (GVSU) 2 semesters, 3 lectures, 2 labs
- Instructor** - CIS-241 System-level Programming and Utilities (GVSU) 5 semesters, 7 sections
- Instructor** - CIS-263 Data Structures and Algorithms (GVSU) 1 semester, 1 section
- Instructor** - CIS-351 Computer Organization (GVSU) 1 semester, 2 sections
- Teaching Assistant** - CSE-491 Secure and Efficient C++ Software Development (MSU) Fall 2023
Co-designed the course with Dr. Charles Ofria. Assisted in all aspects of the course (e.g., creation of course materials, grading, daily class operation, etc.).
- Teaching Assistant** - CSE-431 Algorithm Engineering (MSU) Fall 2022
Assisted Dr. Emily Dolson by holding helproom hours, implementing an autograder in Gradescope, giving oral exams, and responding to student questions online.
- Co-Instructor** - Making a Game of IT (MSU) Summers 2021-2024
Taught an intense week-long Python programming fundamentals course to high school students through a game development lens.
- Research Mentor** - Mentored the following undergraduates at MSU:
- | | |
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| Valdine Peggy Tchinda | 2023 - 2024 |
| Caroline Gormerly (co-mentored with Kate Skocelas) | Fall 2020 |
| Nicholas Lloyd (co-mentored with Alex Lalejini) | Summer 2019 |
| Daniel Junghans (co-mentored with Alex Lalejini) | Summer 2019 |

Mentor - Workshop for Avida-ED Software Development (WAVES) Summers 2020 & 2021
Mentored workshop participants Dylan Rainbow (2020), Lanea Rohan, and Aria Killebrew-Bruehl (2021) in software development for Empirical and MABE2.

Supplemental Instructor - Shawnee State University Springs 2016 & 2017
Assisted professor Jason Witherell with ETGG1803: Concepts of 3D Graphics and Math. Held open lab hours where students could receive additional instruction on concepts and help on assignments.

Outreach & Service

Member - Research Committee, GVSU College of Computing 2025 - Present

Member - CIS 162 & 163 Coordination Committee, GVSU F2025 - W2026
One of four members collaborating to coordinate our intro sequence. Fall 2025 I created a full suite of nine labs in PrairieLearn, worked with Jared Moore to rewrite all four exams, and helped create a repository of in-class activities for CIS 162. Winter 2026 I updated projects and exams.

Leader - Student Success Taskforce, GVSU CS Department 2025 - 2026
Identified college- and department-level initiatives supporting student success.

Faculty Volunteer - Recruitment Events, GVSU College of Computing 2025 - Present
Attended Laker Experience Days, Open Houses, and Computing Spotlights to answer questions from prospective students and their families.

Panelist - Faculty Orientation, GVSU Fall 2025
Invited by FTLC to participate on the junior faculty Q&A panel at faculty orientation

Seminar Organizer - BEACON Center, MSU 2023 - 2024
Initiated, organized, and emceed a bi-weekly seminar on digital evolution.

Volunteer - BEACON Center, MSU 2019 - 2020, 2023
Attended "Science Nights" and Michigan State University's annual Science Festival to teach elementary-aged students about evolution through hands-on activities.

Elected General Chair - Emerging Researchers in Artificial Life 2022 - 2023
Tasked with overseeing all operations of the ERA board and ensuring smooth collaboration among members.

Elected Communications Chair - Emerging Researchers in Artificial Life 2020 - 2021
Managed social media for the group, served as the board's point of contact, and managed file-keeping for chair meetings.

Department Steward - Graduate Employees Union, MSU 2019 - 2020
Served as a main point of communication between the Computer Science & Engineering department and the graduate union at Michigan State University.

Senior Panelist - Gaming Department, SSU 2017 - 2018
Served on a panel answering any questions potential students and their families had about the gaming degree, Shawnee State University, and employment opportunities after graduation.

Volunteer Tutor - 14th Street Community Center, Portsmouth, Ohio Fall 2015
Assisted elementary-aged students with homework understanding and oversaw downtime activities after work was completed.

Reviewer

Evolution Letters	2025 - Present
Genetic and Evolutionary Computation Conference (GECCO)	2025 - Present
Artificial Life Conference	2021 - Present

Honors & Awards

2025 GVSU Student Well-Being Award (Silver)	Winter 2025
2022 ISAL Award for Outstanding Student Publication	Summer 2022
Outstanding Senior in Computer Engineering, SSU	Spring 2018
Outstanding Senior in Digital Simulation and Gaming, SSU	Spring 2018
Outstanding Supplemental Instructor, SSU	Spring 2017
President's List, SSU	Fall 2014 - Spring 2017
Shawnee State Presidential Scholarship, SSU	Fall 2014 - Spring 2018
Margaret Gurney Noel Scholarship, SSU	Fall 2015 - Spring 2016
Centrus Energy Corporation Scholarship, SSU	Fall 2014 - Spring 2016
Ralph and Louise Arrick Scholarship, SSU	Fall 2014 - Spring 2015

Last Updated September 3rd, 2026